A Sink without a Local Source That Is Not Attracting

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Abstract

We answer the local-source converse negatively for finite rational bimatrix games. An explicit five-by-five game has a sink strongly connected component consisting of every pure profile except one corner. Its content has no local source in any product-face subgame, yet fails uniform attraction under the replicator flow. A complete classification of stationary support pairs in a four-by-four block excludes every possible local source, including mixed sources involving the extra strategies. The dynamical obstruction is a backward orbit approaching a six-edge cycle. Exact logarithmic return coordinates give a linear leading map with a strictly positive expanding eigenvector and a uniformly bounded error. An invariant cone supplies the actual orbit and controls it between sections. All payoff entries are rational.

1 The question and the counterexample

For a bimatrix game $A, B \in \mathbb{Q}^{r\times s}$, let $X = \Delta_r \times \Delta_s$, where Δ_j is the probability simplex. Its replicator flow φ_t solves

$$\dot{x}_i = x_i\big((Ay)_i - x^T Ay\big), \qquad \dot{y}_j = y_j\big((x^T B)_j - x^T By\big). \tag{1.1}$$

The polynomial vector field gives a global flow on compact X. Each product face is invariant in both time directions, and a positive coordinate stays positive at every finite time.

The preference graph has the pure profiles as vertices. A directed edge is any unilateral weak payoff improvement between distinct profiles; a payoff tie gives both directed edges. For a sink strongly connected component H, its content is

$$K_H = \{(x, y) : \operatorname{supp}(x) \times \operatorname{supp}(y) \subseteq H\}.$$

A compact invariant set K is uniformly attracting if some relative positively invariant neighborhood U of K satisfies $\sup_{z\in U} \operatorname{dist}(\varphi_t(z), K) \to 0$; this is the uniform convention in [2, Definition 3.2]. The canonical C73-71 question additionally calls a minimal such set an attractor [4].

A local source of H is a point $z \in K_H \cap Y$, where Y is a product-face subgame not contained in K_H, which is a quasi-strict Nash equilibrium of the *negated* game on Y. Equivalently, all supported actions of each player have the same original payoff, and every unsupported action available in Y has strictly larger original payoff. There are no requirements on actions absent from Y. Biggar and Papadimitriou give the local-source obstruction and ask its converse [1, Definition 2.1, Lemma 2.2, Section 4].

Set $\eta = 1/100$. Label the rows $1, 2, 3, h, r$ and columns $1, 2, 3, d, c$, in that order, and define

$$A = \begin{pmatrix} 2 & 0 & -1 & 1 & 0 \\ -1 & 2 & 0 & 1 & 0 \\ 0 & -1 & 2 & 1 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ -\eta & -\eta & -\eta & -\eta & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & -1 & 0 & -1/2 & -\eta \\ 0 & 2 & -1 & -1/2 & -\eta \\ -1 & 0 & 2 & -1/2 & -\eta \\ 1 & 1 & 1 & 2 & -\eta \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix}. \tag{1.2}$$

Theorem 1. *In* (1.2)*, let H be all 25 pure profiles except (r, c). Then H is one sink strongly connected component,*

$$K_H = \{(x, y) : x_r y_c = 0\},$$

and H has no local source in any product-face subgame. Nevertheless, K_H is not uniformly attracting, and therefore is not an attractor. This disproves the canonical local-source converse.

The earlier catalogue attempts prove positive results for rectangles, crosses and two-by-two games [5, 6]; those results do not settle this converse. Our six-edge set is not itself the sink component of the full game, so the theorem on simple sink cycles in [3] does not apply to it.

2 The sink and every local-source candidate

Call rows $1, 2, 3, h$ and columns $1, 2, 3, d$ the resident block. The old core on indices $1, 2, 3$ has matrices

$$A_0 = \begin{pmatrix} 2 & 0 & -1 \\ -1 & 2 & 0 \\ 0 & -1 & 2 \end{pmatrix}, \qquad B_0 = A_0^T.$$

A *stationary support pair* has equal payoffs on each player's positive support; unused actions are unrestricted. Thus it need not be a Nash equilibrium of the full game.

Lemma 2. *The set H in Theorem 1 is a sink strongly connected component, and its content is $\{x_r y_c = 0\}$.*

Proof. Read old indices cyclically modulo three. Write $F_i = (i, i+1)$ and $G_i = (i, i-1)$; their payoffs are respectively $(0, -1)$ and $(-1, 0)$. The strict improvements $F_i \to G_i \to F_{i+1}$ form a six-cycle. Each old diagonal is reachable from that cycle by an improvement to payoff 2. From a diagonal there is a path

$$(i, i) \longrightarrow (h, i) \longrightarrow (h, d) \longrightarrow (j, d) \longrightarrow G_j,$$

whose moving-player payoffs improve respectively from 2 to 3, from 1 to 2, from 0 to 1, and from $-1/2$ to 0. These paths also contain every remaining resident vertex, so the resident block is strongly connected.

All (r, j) with resident j communicate through column-payoff ties. Each can move to (h, j), whose row payoff is at least $0 > -\eta$. Conversely, an old profile with row payoff -1 can move to row r. All (i, c) with resident i likewise communicate through row-payoff ties; an old column payoff -1 allows entry, and a resident column payoff at least zero allows exit. This proves strong connectivity of all of H.

The only possible outgoing edges would move from an arm to (r, c). Both such moves lower the moving player's payoff from 0 to -1, so no weak outgoing edge exists. Finally a product distribution gives positive mass to the sole omitted profile exactly when $x_r y_c > 0$. The stated content follows; it is a compact invariant union of two faces. □

Lemma 3. *Every stationary support pair of the old core is either pure, a principal two-by-two pair with both payoffs* $4/5$*, or the fully mixed uniform pair with both payoffs* $1/3$*.*

Proof. Against a pure action, the other player's payoffs are the three distinct numbers $2, 0, -1$; hence one-by-two and one-by-three supports, and their reversals, are impossible. If one player uses all three actions, indifference takes the form $A_0 z = v\mathbf{1}$. Since $\det A_0 = 7$ and $A_0\mathbf{1} = \mathbf{1}$, normalization forces $z = (1/3, 1/3, 1/3)$ and $v = 1/3$. This excludes three-by-two and two-by-three supports and identifies the full-support pair.

For two-by-two supports, cyclic relabelling reduces the row support to $\{1, 2\}$. On column support $\{1, 3\}$, column 1 strictly dominates column 3 for these rows. On column support $\{2, 3\}$, row 2 strictly dominates row 1. Only the principal support remains. There both indifference equations give masses $2/5, 3/5$ on $i, i+1$, and payoff $4/5$. The pure profiles complete the list. □

Lemma 4 (Resident sign certificate)**.** *Every stationary support pair of the resident four-by-four block has at least one nonnegative payoff.*

Proof. When neither h nor d is supported, Lemma 3 applies. Its pure payoffs are $(2, 2)$, $(0, -1)$, and $(-1, 0)$, so the claim holds in that case.

Suppose d is supported and h is not. Column d earns $-1/2$. Two old columns cannot both tie it: the three coordinates of $A_0 x$ sum to 1, so two coordinates equal to $-1/2$ would force the third to be 2. Attaining that maximum forces x to be the corresponding pure action, whose other column payoffs are 0 and -1, a contradiction. If just one old column joins d, all old row payoffs are distinct: they are a positive multiple of $2, 0, -1$, plus a common constant. Thus the row must be pure, but then its old-column payoff cannot equal $-1/2$. The only remaining case is the pure column d with any old-row mixture, giving payoffs $(1, -1/2)$.

If h is supported and d is not, row h earns 3, whereas each old row earns at most 2. Therefore only h is played, with payoffs $(3, 1)$.

Finally suppose both hub actions are supported. If either old support is empty, indifference forces both to be empty, giving (h, d) and payoffs $(0, 2)$. Otherwise let $\alpha, \beta \in (0, 1)$ be the old-column and old-row masses, and normalize their old mixtures to $\bar{y}, \bar{x}$. Indifference to the hubs gives

$$(A_0\bar{y})_i = 4 - \frac{1}{\alpha} =: v, \qquad (A_0\bar{x})_j = \frac{1}{\beta} - \frac{3}{2} =: w$$

on the respective old supports. Thus $(\bar{x}, \bar{y})$ is one of the core pairs in Lemma 3, and

$$\alpha = \frac{1}{4 - v}, \qquad \beta = \frac{2}{2w + 3}, \qquad (U_A, U_B) = \left(\frac{3}{4 - v}, \frac{4w + 1}{2w + 3}\right).$$

The condition $\beta < 1$ requires $w > -1/2$, excluding the only core value $w = -1$. All other core pairs have $w \geq 0$ and $v < 3$, so these two resident payoffs are positive. This exhausts the support cases. □

Proposition 5. *No point of* K_H *is a local source in any product-face subgame.*

Proof. A product-face subgame is not contained in K_H exactly when it includes both row r and column c. Consider any proposed source in such a subgame. It must be stationary on its positive support.

If $x_r = y_c = 0$, it is a resident stationary pair. Quasi-strictness in the negated game would require both resident payoffs to be strictly less than $-\eta$, because the added actions r, c each earn $-\eta$. Lemma 4 excludes this.

Suppose $x_r > 0, y_c = 0$. If only row r is played, supported columns earn 0, whereas column c earns -1; hence the negated-game Nash condition fails. Otherwise normalize the resident row mass $1 - x_r$. Row indifference to r says that its resident payoff is $-\eta < 0$. Since row r contributes zero to all resident columns, their indifference equations give a resident stationary pair. By Lemma 4 its column payoff v_B is nonnegative. In the full profile, supported columns earn $(1-x_r)v_B \geq 0$, whereas adding c earns $-\eta(1-x_r) - x_r < 0$. This also violates negated-game Nash.

If $x_r = 0, y_c > 0$, the symmetric normalization argument uses the resident column payoff $-\eta$, and therefore a nonnegative resident row payoff v_A. Supported rows earn $(1-y_c)v_A \geq 0$, whereas adding r earns $-\eta(1-y_c) - y_c < 0$. If only column c is played, the same contradiction is directly $0 > -1$. The remaining case $x_r > 0, y_c > 0$ is outside K_H. These cases cover every support and every subgame, including omitted resident actions; no condition on such omitted actions has been imposed. □

3 An orbit approaching the cycle in negative time

Set the hub probabilities to zero. Face invariance leaves the four-by-four game on rows $1, 2, 3, r$ and columns $1, 2, 3, c$,

$$A_* = \begin{pmatrix} 2 & 0 & -1 & 0 \\ -1 & 2 & 0 & 0 \\ 0 & -1 & 2 & 0 \\ -\eta & -\eta & -\eta & -1 \end{pmatrix}, \qquad B_* = A_*^T. \tag{3.1}$$

Write ψ_t for the replicator flow with payoffs $-A_*, -B_*$. Negating both payoffs negates the vector field, so $\psi_t = \varphi_{-t}$ on this face. Let S be the union of the six edges in the cyclic list

$$(2,3), (1,3), (1,2), (3,2), (3,1), (2,1), (2,3). \tag{3.2}$$

Every vertex and every edge of S is contained in K_H.

Lemma 6. *There is a point z_* with all eight probabilities positive in* (3.1) *such that* $\operatorname{dist}(\psi_t(z_*), S) \to 0$ *as* $t \to \infty$.

The rest of this section proves the lemma directly from (1.1). All constants used below are independent of the scale of the logarithmic coordinates.

3.1 Coordinates and the linear leading map

Use an incoming section $x_1 = x_2$ with column 3 dominant. Define

$$u = \log \frac{x_1}{x_3}, \quad v = \log \frac{y_3}{y_2}, \quad w = \log \frac{y_3}{y_1}, \quad d = \log \frac{x_1}{x_r}, \quad e = \log \frac{y_3}{y_c}. \tag{3.3}$$

Every positive vector of these gaps determines unique strictly positive probabilities on the section:

$$x_1 = x_2 = \frac{1}{2 + \exp(-u) + \exp(-d)}, \quad x_3 = \exp(-u)x_1, \quad x_r = \exp(-d)x_1,$$

$$y_3 = \frac{1}{1 + \exp(-v) + \exp(-w) + \exp(-e)},$$

$$y_2 = \exp(-v)y_3, \quad y_1 = \exp(-w)y_3, \quad y_c = \exp(-e)y_3.$$

For frozen play at $(1,3)$, the next column tie occurs at time v. At that time the unused gaps would be $u + 3v, w + 2v, d + (1-\eta)v, e - \eta v$, and $\log(x_1/x_2) = v$. Exchange the players, then

relabel the old indices simultaneously by $2 \mapsto 1, 3 \mapsto 2, 1 \mapsto 3$, exchanging r, c too. Game (3.1) is invariant under this operation. It turns the new tie into the same incoming section. The resulting leading map is

$$M \begin{pmatrix} u \\ v \\ w \\ d \\ e \end{pmatrix} = \begin{pmatrix} 2v + w \\ u + 3v \\ v \\ e - \eta v \\ d + (1 - \eta)v \end{pmatrix}. \tag{3.4}$$

Its resident block is

$$L = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 3 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad p(z) = \det(zI - L) = z^3 - 3z^2 - 2z - 1.$$

There is a unique positive root $\lambda \in (3, 4)$: the function $3/t + 2/t^2 + 1/t^3$ strictly decreases from infinity to zero, and $p(3) < 0 < p(4)$. It is simple since $p'(\lambda) > 0$. If z is any root, then $|z|^3 \le 3|z|^2 + 2|z| + 1$, so $|z| \le \lambda$. Equality in the triangle inequality would force $z^2, z, 1$ to have the same argument, and hence $z = \lambda$. Thus every other root has strictly smaller modulus. The remaining block of M is the swap matrix, with eigenvalues $1, -1$. Consequently λ is a simple strictly dominant eigenvalue of M.

Normalize the second coordinate to one. A corresponding eigenvector is

$$q = \left(\lambda - 3, 1, \lambda^{-1}, D, E\right), \qquad D = \frac{1 - \eta(\lambda + 1)}{\lambda^2 - 1}, \qquad E = \lambda D + \eta. \tag{3.5}$$

All coordinates are positive, and $E - \eta = \lambda D > 0$, because $\eta = 1/100 < 1/(\lambda + 1)$.

3.2 A uniform error bound for the actual flow

Lemma 7. *On a sufficiently narrow closed cone about the ray $\mathbb{R}_{>0}q$, for all sufficiently large gap vectors $z = (u, v, w, d, e)$, the first subsequent tie $y_2 = y_3$ exists at time $T = v + O(1)$. After the relabeling above, the actual return map satisfies*

$$P(z) = Mz + O(1) \tag{3.6}$$

uniformly on that cone.

Proof. By (3.5), choose the cone sufficiently narrow that, for some fixed $\mu > 0$, the four affine functions

$$u + 3t, \quad w + 2t, \quad d + (1 - \eta)t, \quad e - \eta t \tag{3.7}$$

are at least $4\mu v$ for $0 \le t \le v$. The normalized section $v = 1$ of the cone is compact. Thus all initial coordinates are also at most a fixed multiple of v.

Stop at the first of the following: the tie $y_2 = y_3$, the time $v + 10$, or an unused gap $u(t), w(t), d(t), e(t)$ reaching $2\mu v$. Before this stopping time, each unused probability x_3, x_r, y_1, y_c is at most $\exp(-2\mu v)$. Put $G(t) = \log(x_1/x_2)$ and $D_0(t) = \log(y_3/y_2)$. Exact payoff differences give

$$G' = y_3 + 2y_2 - 3y_1 \ge 1 - 4y_1 - y_c \ge \tfrac{1}{2}, \qquad G(0) = 0, \tag{3.8}$$

$$D_0' = -x_1 + 3x_2 - 2x_3 = -1 + 4x_2 - x_3 + x_r, \qquad D_0(0) = v. \tag{3.9}$$

The last inequality in (3.8) holds for large v. Therefore $x_2 \le e^{-t/2}$, and integration of (3.9) shows, throughout the stopped passage,

$$|D_0(t) - (v - t)| \le 8 + 2(v + 10)e^{-2\mu v} \le 9 \tag{3.10}$$

for sufficiently large v. There is no tie before $v-9$. If the other stopping conditions do not occur first, a tie must occur by $v+10$, since otherwise (3.10) would make its strictly positive gap negative.

These estimates also bound the integrated deviations from the frozen vertex, up to any such stopping time τ:

$$\int_0^\tau x_2(t)\,dt \le 2, \qquad \int_0^\tau y_2(t)\,dt \le e^9 \int_0^{v+10} e^{-v+t}\,dt \le e^{19}. \tag{3.11}$$

Here $y_2 \le e^{-D_0}$ was used. Each unused probability has integral at most $(v+10)e^{-2\mu v}$. The four exact unused-gap derivatives are

$$\begin{aligned} u' &= 3 - 4y_2 - 5y_1 - 3y_c, \\ d' &= (1-\eta) - y_2 - 3y_1 - (2-\eta)y_c, \\ w' &= 2 - x_2 - 5x_3 - 2x_r, \\ e' &= -\eta + x_2 - 2x_3 + (\eta - 1)x_r. \end{aligned} \tag{3.12}$$

By (3.11), their integrals differ from (3.7) by a fixed constant, independent of v. Up to $v+10$, the ideal lower bound $4\mu v$ decreases by at most a fixed constant. For sufficiently large v, the actual gaps therefore remain strictly above $2\mu v$. This rules out a first bootstrap failure without assuming that the passage has already completed.

The tie must consequently occur with $v-9 \le T \le v+10$. Near that window, (3.9) gives $D_0' < -1/2$, so it is the unique transverse crossing there. Equations (3.8) and (3.11) yield $G(T) = v + O(1)$, while (3.12) yields the other four affine updates with bounded errors. The relabelled gap vector is exactly $(w(T), u(T), G(T), e(T), d(T))$, proving (3.6). □

3.3 An invariant cone and the complete orbit

Let ℓ be a left λ-eigenvector of M, normalized by $\ell(q) = 1$. One may take a positive multiple of $(1, \lambda, 1/\lambda, 0, 0)$. Write $z = \tau q + h$, with $\tau = \ell(z)$ and $h \in \ker \ell$. The restriction $N = M|_{\ker \ell}$ has spectral radius strictly less than λ. Choose s between these two numbers. There is a norm with $\|Nh\|_* \le s\|h\|_*$; explicitly,

$$\|h\|_* := \sum_{k=0}^{\infty} s^{-k} \|N^k h\|_\infty$$

works. The series converges by the strict spectral-radius bound (for example by Jordan form), and shifting its index proves the stated inequality.

Choose $\delta > 0$ sufficiently small that

$$\mathcal{C} = \{\tau q + h : \tau > 0,\ \ell(h) = 0,\ \|h\|_* \le \delta\tau\}$$

lies in the cone of Lemma 7. All gap coordinates on $\mathcal{C}$ are positive and comparable with τ. The bounded return error gives fixed constants C_1, C_2 such that

$$\tau_{\text{new}} \ge \lambda\tau - C_1, \qquad \|h_{\text{new}}\|_* \le s\delta\tau + C_2.$$

For sufficiently large τ,

$$(\lambda - s)\delta\tau > C_2 + \delta C_1, \qquad \tau_{\text{new}} \ge \frac{\lambda+1}{2}\tau.$$

Thus the high-scale part of $\mathcal{C}$ is invariant under the actual return map, and the scales grow geometrically. A sufficiently large multiple of q gives a nonempty initial section through (3.3).

Each successive passage exists by Lemma 7; its duration is $v_j + O(1)$, so the sum of passage times tends to infinity.

It remains to control the trajectory between sections. The unused probabilities tend uniformly to zero in each passage. For $t \le v_j/2$, (3.10) gives $y_2 \le e^{-v_j/2+9}$; for $t \ge v_j/2$, (3.8) gives $x_2 \le e^{-v_j/4}$. In the first interval, setting the small unused probabilities and y_2 to zero and normalizing gives a point on the incoming edge $\{1,2\} \times \{3\}$. In the second interval, doing the same with x_2 gives a point on the outgoing edge $\{1\} \times \{3,2\}$. Normalization changes the state by a quantity tending uniformly to zero. Both edges belong to S. The finite symmetry relabelings preserve S; hence the same estimate applies at every phase. Since $v_j \to \infty$, the entire trajectory approaches S, proving Lemma 6.

4 Completion and the obstruction

Proof of Theorem 1. Lemma 2 identifies the sink and its content, and Proposition 5 excludes all local sources. Choose the point z_* supplied by Lemma 6, and set its two hub coordinates to zero in the full game. Its probabilities on the four-by-four face are positive, so $x_r y_c > 0$ and $z_* \notin K_H$. By compactness, $d_* := \operatorname{dist}(z_*, K_H) > 0$.

Face invariance and exact time reversal give

$$\operatorname{dist}(\varphi_{-t}(z_*), K_H) \longrightarrow 0.$$

Every relative neighborhood U of compact K_H contains a sufficiently thin metric neighborhood of K_H. Therefore $\varphi_{-t}(z_*) \in U$ for all sufficiently large t. The flow identity then implies

$$\sup_{z \in U} \operatorname{dist}(\varphi_t(z), K_H) \;\ge\; \operatorname{dist}(\varphi_t(\varphi_{-t}(z_*)), K_H) \;= d_* > 0.$$

Uniform attraction fails for every neighborhood, including every positively invariant one. This proves the theorem. □

The obstruction is an orbit that is asymptotic to a resident cycle in backward time while both extra strategies remain positive at finite times. Their favorable and unfavorable phases have geometrically different durations. The positive eigenvector in (3.5), together with the actual-flow error bound, controls those durations and both extra coordinates. No stationary point is used as a local source. The hub actions make all but one corner a single sink without changing this invariant-face orbit.

Remark 8 (Scope)**.** This is a counterexample to the entire main bimatrix implication, under its weak-edge, all-subgame and uniform-attraction conventions. It also refutes the corresponding unrestricted many-player implication by adding players with one action. It does not establish failure of the separately stated, weaker condition that every full-interior initial state converges pointwise to K_H in forward time: the present witness lies on a product face and uses time-shifted initial states. An orbit starting in invariant K_H never leaves it.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

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